UNIT 3 · TOPIC 12(A)

Measurement of Engine Power

Syllabus topic: Measurement of engine power; electric motors and their characteristics including batteries

Power terms

$$\text{Indicated power IP (kW)} = p_m \cdot L \cdot A \cdot n \cdot K / 60{,}000$$

IP is developed inside the cylinder on the piston (from the indicator diagram). p_m = mean effective pressure (kPa); L = stroke (m); A = piston area (m²); n = power strokes/min (= rpm/2 for 4-stroke, = rpm for 2-stroke); K = number of cylinders.

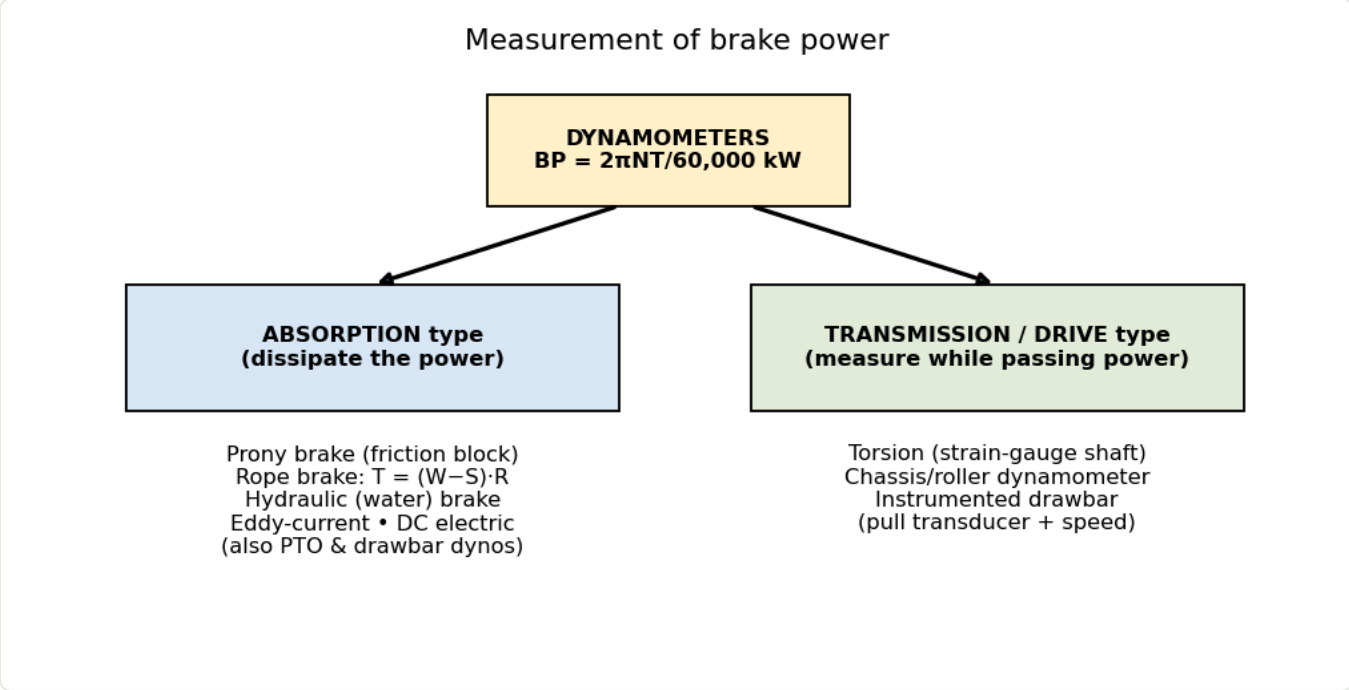
$$\text{Brake power BP (kW)} = 2\pi NT/60{,}000 \ (N = \text{rpm}, T = \text{torque in N·m})$$

$$\text{Friction power } FP = IP - BP; \text{ mechanical efficiency } \eta_m = BP/IP$$

- BP = usable power at the flywheel/crankshaft. η_m is typically **75–85%** at rated load.
- Downstream: PTO power ≈ **87–92%** of BP; drawbar power is lower still (Unit 2, Topic 5).

Worked BP example. T = 200 N·m at 2000 rpm → BP = 2π × 2000 × 200/60,000 = **41.9 kW** (or T × N/9550).

Dynamometers



Measurement of brake power — absorption and transmission dynamometers

- **Rope brake:** rope on a flywheel of effective radius R with dead load W and spring balance S → T = (W − S)·R; BP = 2πN(W − S)R/60,000.
- **Hydraulic and eddy-current dynamometers** are standard for tractor PTO tests.
- **Strain-gauge torsion meters** measure transmitted torque without absorbing it.

- **Drawbar power:** instrumented drawbar (pull transducer) + measured speed.

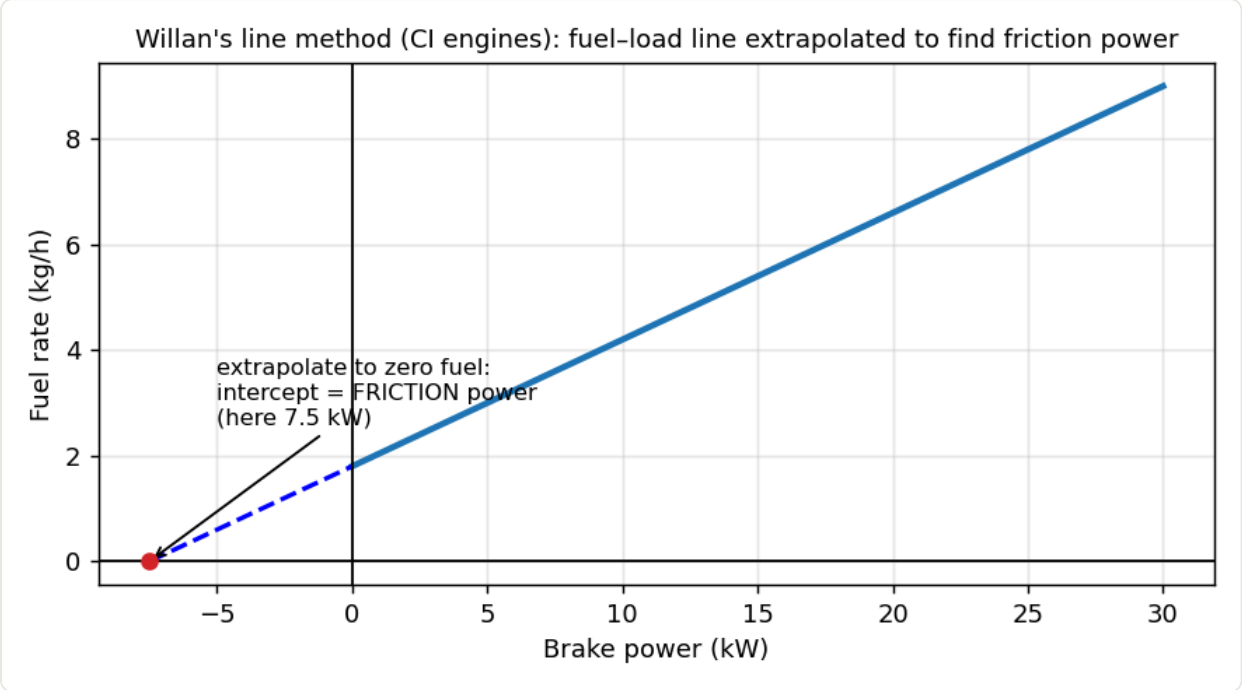
Friction power: Morse test, Willan's line, motoring

Morse test (multi-cylinder engines):

1. Run the engine at constant speed.
2. Cut out one cylinder at a time and record the BP.
3. The drop in BP = the indicated power of that cylinder.
4. Sum the per-cylinder IP; FP = IP − BP.

> **Example.** 4-cyl engine, BP = 40 kW; with cylinders cut out, BP reads 28, 28.5, 28, 27.5 kW → per-cylinder IP = 12, 11.5, 12, 12.5 → total IP = **48 kW** → FP = **8 kW**, η_m = 40/48 = **83.3%**.

- **Willan's line** (diesel only): plot fuel rate vs BP at constant speed — nearly straight; extrapolate to zero fuel. The negative intercept = friction power.
- **Motoring test:** drive the hot engine externally and measure the power absorbed.
- **Retardation/deceleration test:** uses the flywheel slow-down rate.



Willan's line method (CI engines)

Fuel-based indices

$$\text{SFC} = \text{fuel (g or kg) per kWh};\ \eta_{bt} = 3600 \div [\text{SFC (kg/kWh)} \times \text{CV (kJ/kg)}]$$

> **Example.** SFC 0.25 kg/kWh, CV 42,500 kJ/kg → η_bt = 3600/10,625 = **33.9%**.

- Good tractor diesels: SFC ≈ **240–260 g/kWh** at rated load.

Practice MCQs

Q141. Brake power in kW from torque T (N·m) at N rpm is:

(a) NT/60
(b) 2πNT/60,000
(c) πNT/30,000×2
(d) NT/9550 only in W

Q142. T = 200 N·m at 2000 rpm gives BP ≈

(a) 21 kW
(b) 33.5 kW
(c) 41.9 kW
(d) 62.8 kW

Q143. For a 4-stroke engine, power strokes per minute equal:

(a) rpm
(b) rpm/2
(c) 2×rpm
(d) rpm/4

Q144. Mechanical efficiency of an engine is:

(a) IP/BP
(b) BP/IP
(c) FP/IP
(d) BP×IP

Q145. Friction power equals:

(a) BP−IP
(b) IP−BP
(c) IP+BP
(d) BP/IP

Q146. The prony brake and rope brake are examples of:

(a) Transmission dynamometers
(b) Absorption dynamometers
(c) Chassis dynos
(d) Flow meters

Q147. In a rope brake with dead load W, spring balance S and effective radius R, torque =

(a) (W+S)R
(b) (W−S)R
(c) WR/S
(d) SR−W

Q148. Tractor PTO power tests commonly use:

(a) Rope brake
(b) Hydraulic or eddy-current dynamometer
(c) Spring balance
(d) Prony block

Q149. A torsion (strain-gauge shaft) dynamometer:

(a) Absorbs all power
(b) Measures torque while transmitting power
(c) Measures fuel
(d) Measures slip

Q150. The Morse test is applicable to:

(a) Single-cylinder engines
(b) Multi-cylinder engines
(c) Electric motors
(d) Batteries

Q151. 4-cyl engine BP = 40 kW; cut-out readings 28, 28.5, 28, 27.5 kW. Indicated power =

(a) 44
(b) 46
(c) 48
(d) 52 kW

Q152. Its mechanical efficiency ≈

(a) 75%
(b) 80%
(c) 83.3%
(d) 90%

Q153. Willan's line method estimates friction power for:

(a) SI engines only
(b) CI (diesel) engines at constant speed
(c) Motors
(d) Gas turbines

Q154. In Willan's method, FP is read from:

(a) Slope of the line
(b) Extrapolated negative intercept on the power axis at zero fuel
(c) Peak point
(d) Area under curve

Q155. SFC 0.25 kg/kWh with CV 42,500 kJ/kg gives brake thermal efficiency ≈

(a) 24%
(b) 30%
(c) 34%
(d) 42%

Q156. A good tractor diesel at rated load has SFC of about:

(a) 150
(b) 250
(c) 400
(d) 600 g/kWh

Answers

Q141 — (b) 2πNT/60,000
Q142 — (c) 41.9 kW — *BP = 2π×2000×200/60,000 = 41.89 kW (or T×N/9550 = 200×2000/9550).*
Q143 — (b) rpm/2
Q144 — (b) BP/IP
Q145 — (b) IP-BP
Q146 — (b) Absorption dynamometers
Q147 — (b) (W-S)R
Q148 — (b) Hydraulic or eddy-current dynamometer
Q149 — (b) Measures torque while transmitting power
Q150 — (b) Multi-cylinder engines
Q151 — (c) 48 — *Per-cylinder IP = 40-28 = 12, 11.5, 12, 12.5 → ΣIP = 48 kW.*
Q152 — (c) 83.3% — *η_m = 40/48 = 83.3%; FP = 8 kW.*
Q153 — (b) CI (diesel) engines at constant speed
Q154 — (b) Extrapolated negative intercept on the power axis at zero fuel
Q155 — (c) 34% — *η_bt = 3600/(0.25×42500) = 33.9%.*
Q156 — (b) 250

Sources: Tractors and their Power Units (Liljedahl et al.); Kepner et al.; BIS publications; ISO/OECD/ASABE documentation; NPTEL lectures on farm power and IC engines.